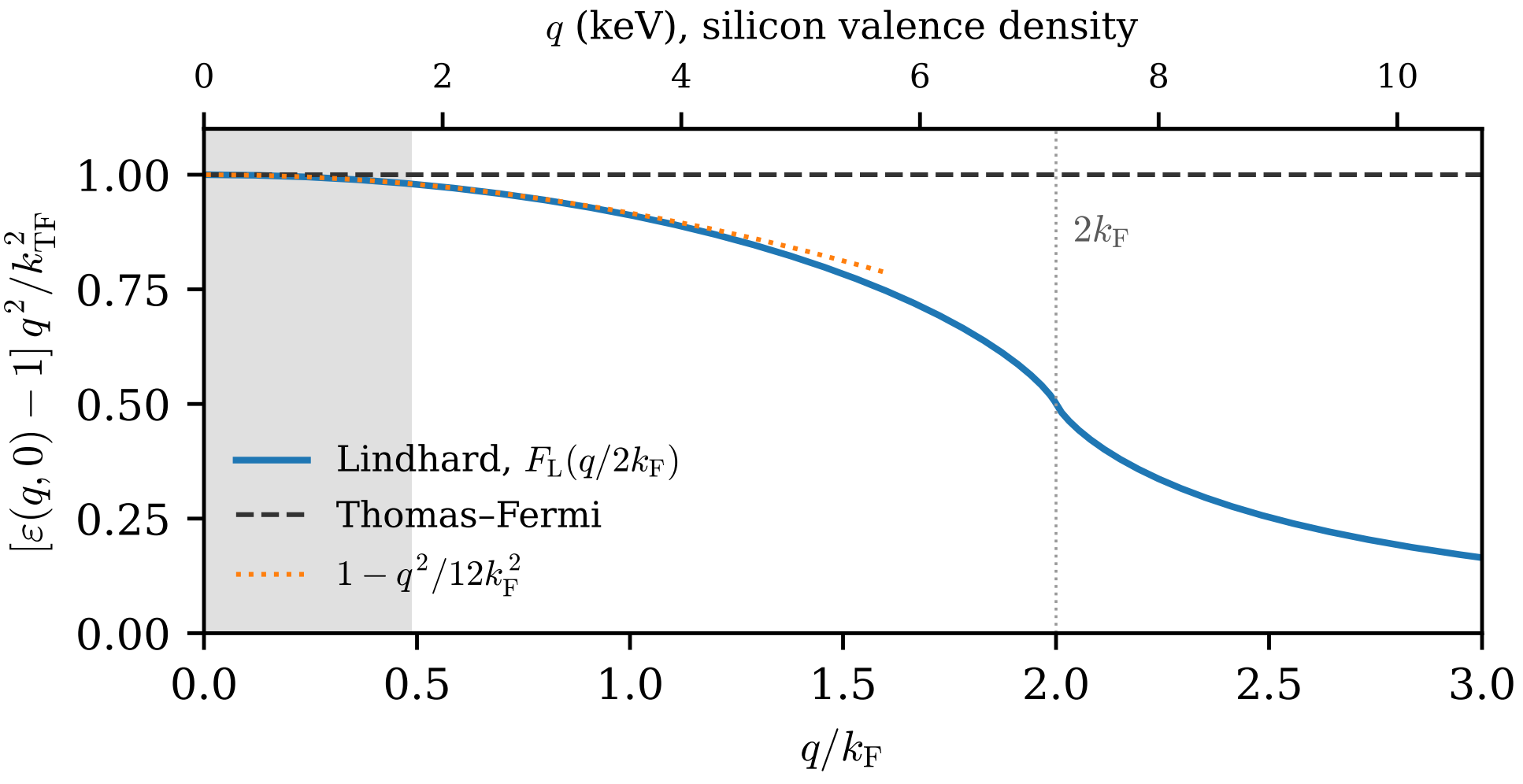
$q$ (keV), silicon valence density
0
2
4
6
8
10
1.00
0.75
0.50
0.25
0.00
$[\varepsilon(q,0)-1]\,q^2/k_{\mathrm{TF}}^2$
$2k_{\mathrm{F}}$
Lindhard, $F_{\mathrm{L}}(q/2k_{\mathrm{F}})$
Thomas–Fermi
$1-q^2/12k_{\mathrm{F}}^2$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
$q/k_{\mathrm{F}}$